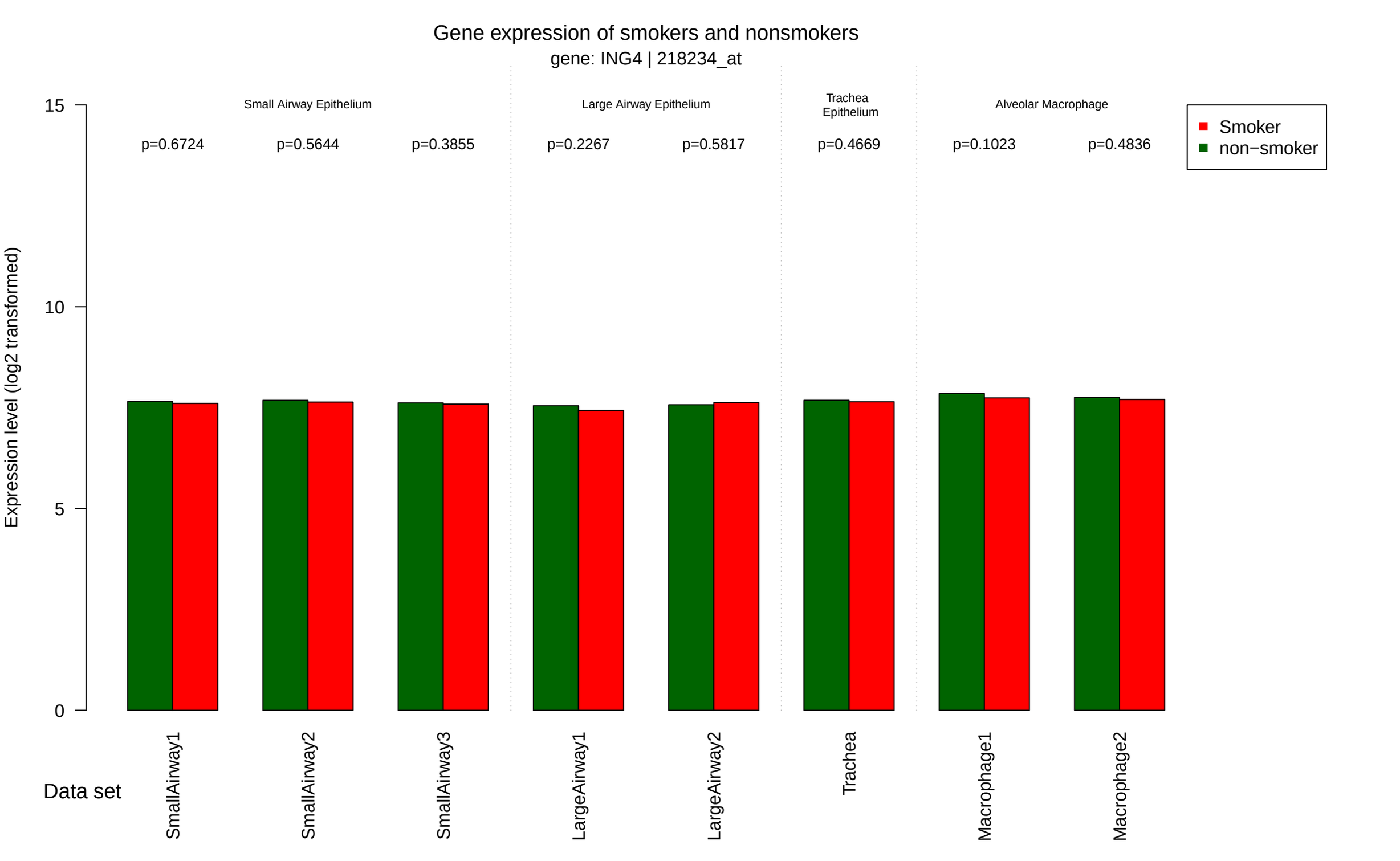
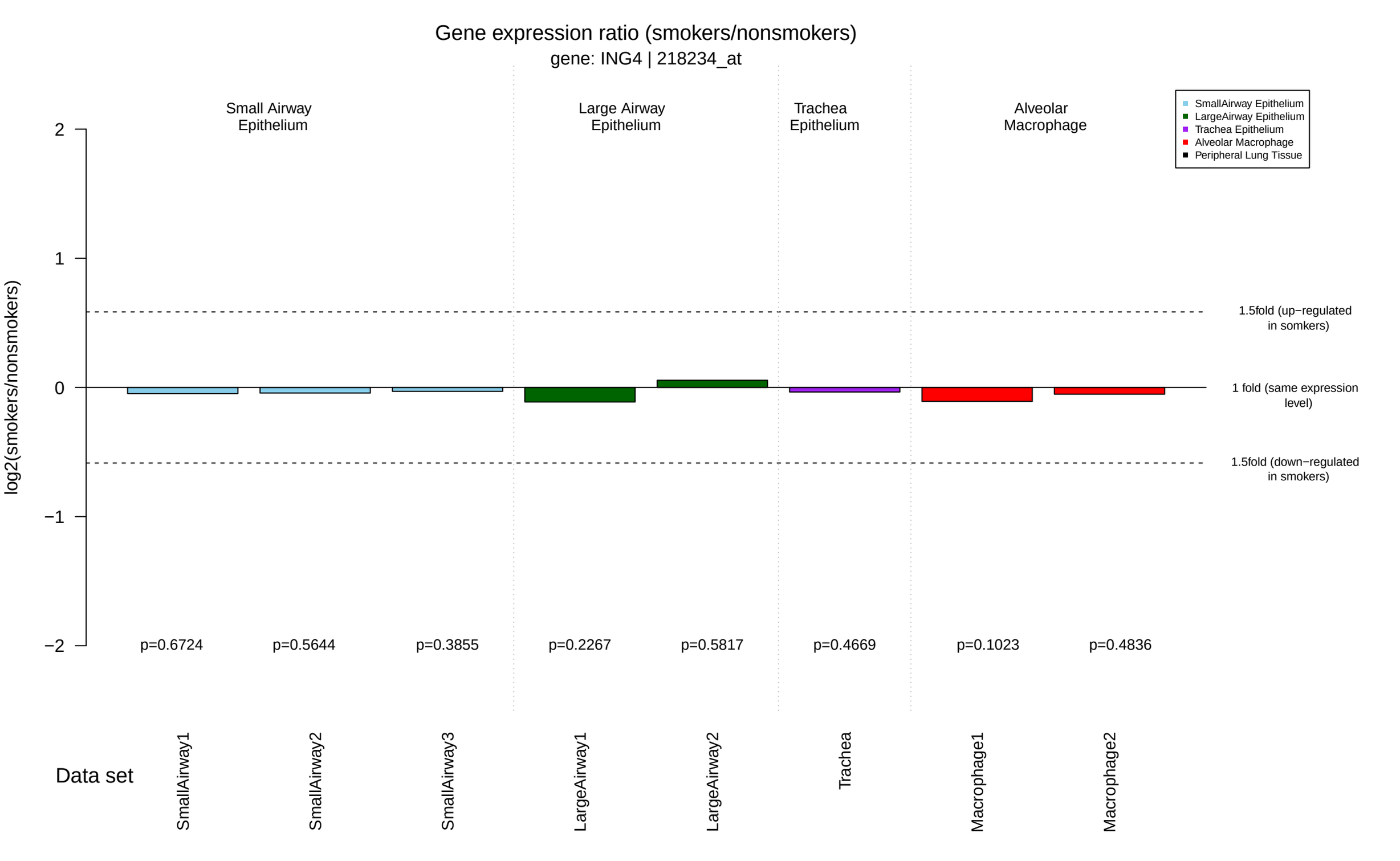
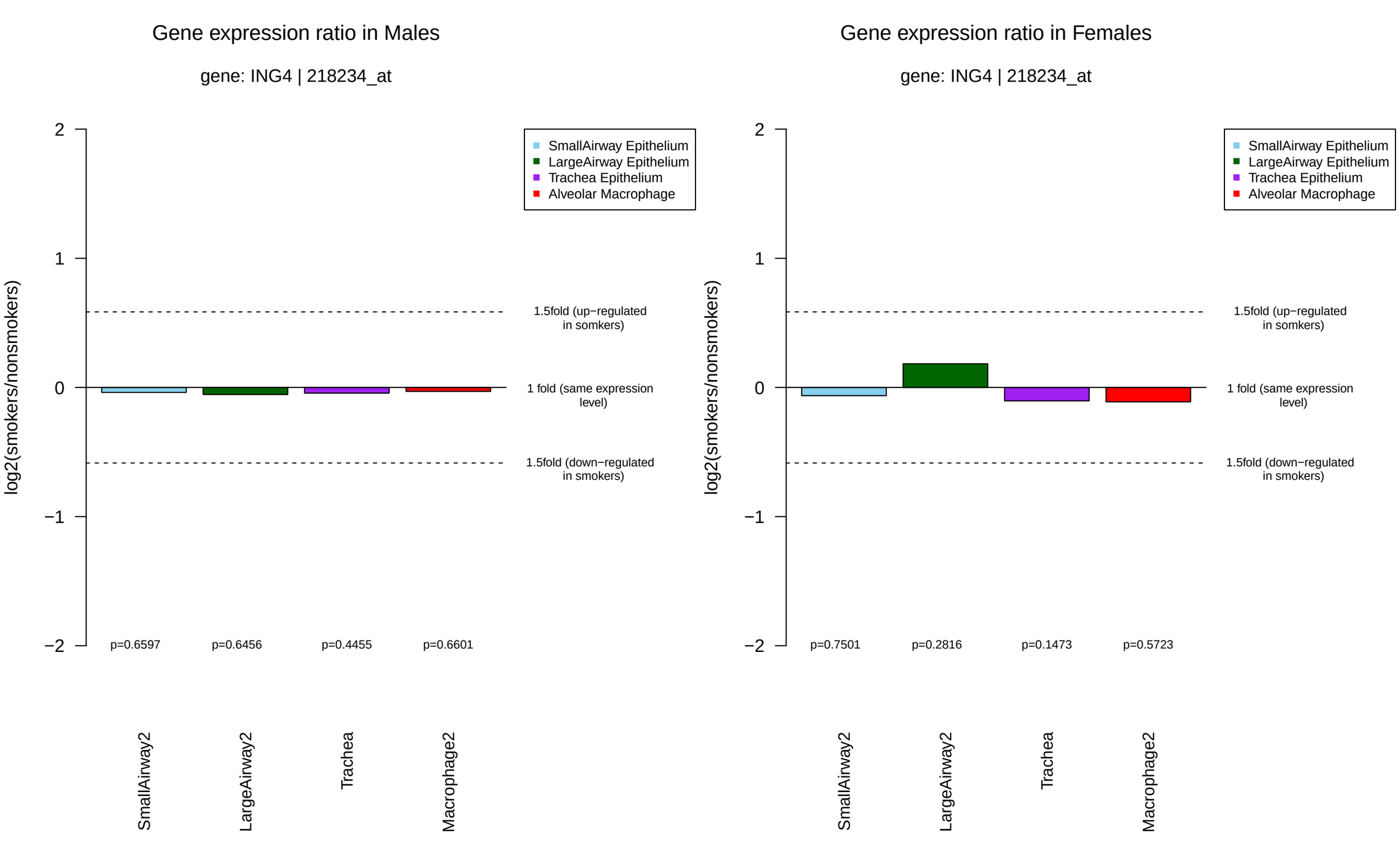
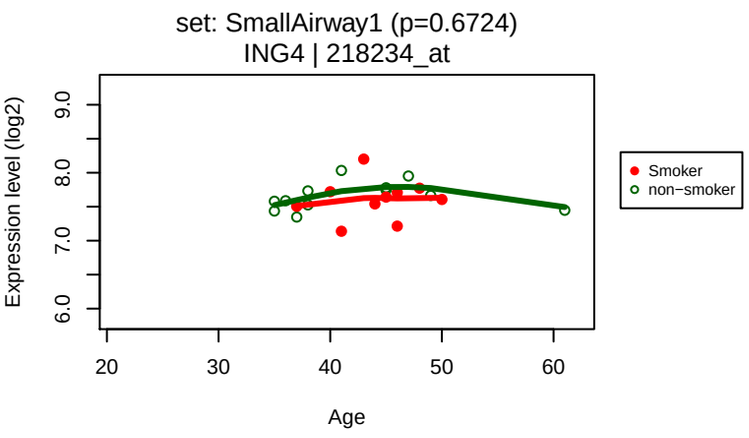
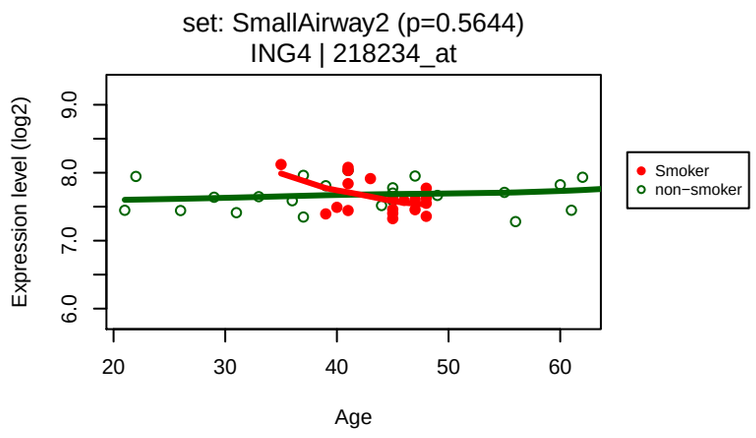
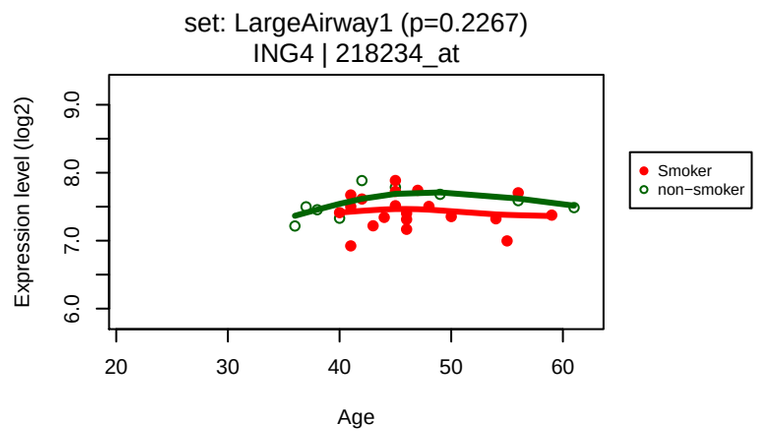

Gene expression of smokers and nonsmokers
gene: ING4 | 218234_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.6724
p=0.5644
p=0.3855
p=0.2267
p=0.5817
p=0.4669
p=0.1023
p=0.4836
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

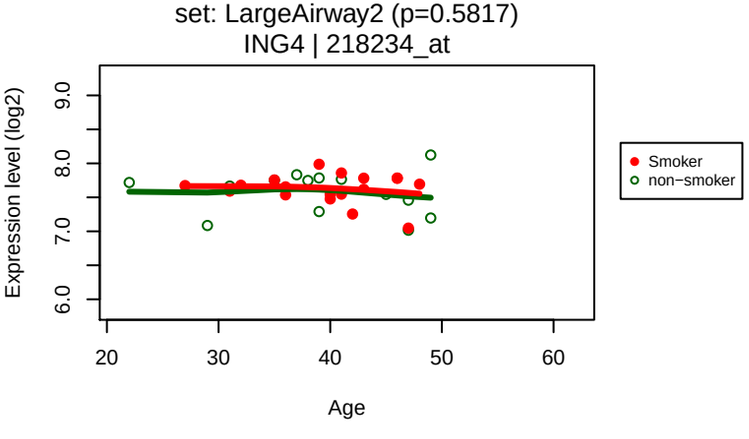

Gene expression ratio (smokers/nonsmokers)
gene: ING4 | 218234_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6724
p=0.5644
p=0.3855
p=0.2267
p=0.5817
p=0.4669
p=0.1023
p=0.4836
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

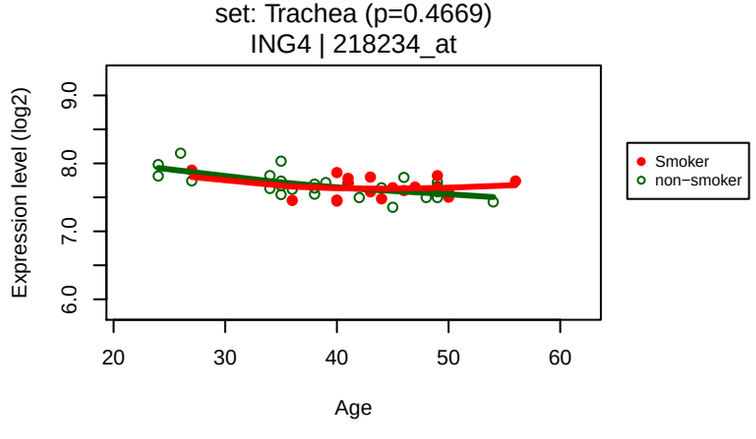

Gene expression ratio in Males
gene: ING4 | 218234_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6597
p=0.6456
p=0.4455
p=0.6601
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ING4 | 218234_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7501
p=0.2816
p=0.1473
p=0.5723
SmallAirway2
LargeAirway2
Trachea
Macrophage2

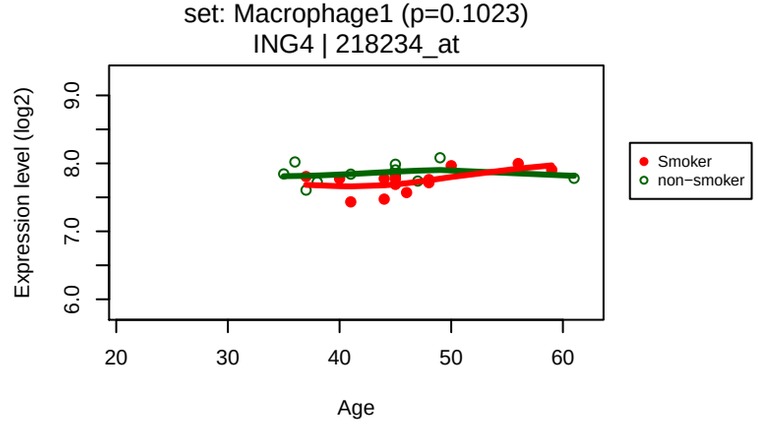

set: SmallAirway1 (p=0.6724)
ING4 | 218234_at
Expression level (log2)
6.0
7.0
8.0
9.0
20
30
40
50
60
Age
Smoker
non−smoker

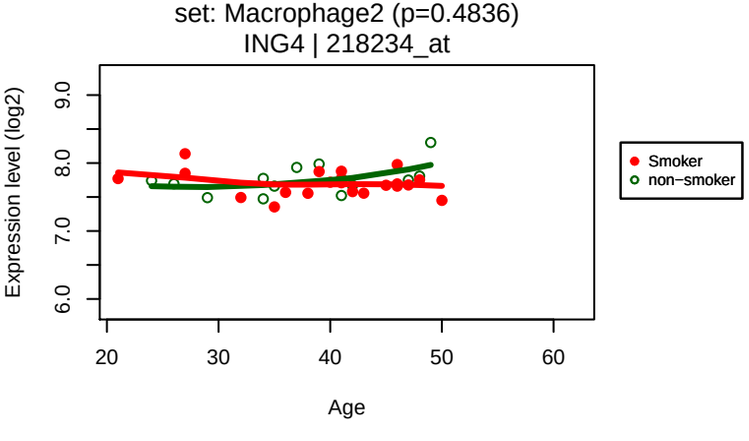

set: SmallAirway2 (p=0.5644)
ING4 | 218234_at
Expression level (log2)
6.0
7.0
8.0
9.0
20
30
40
50
60
Age
Smoker
non−smoker

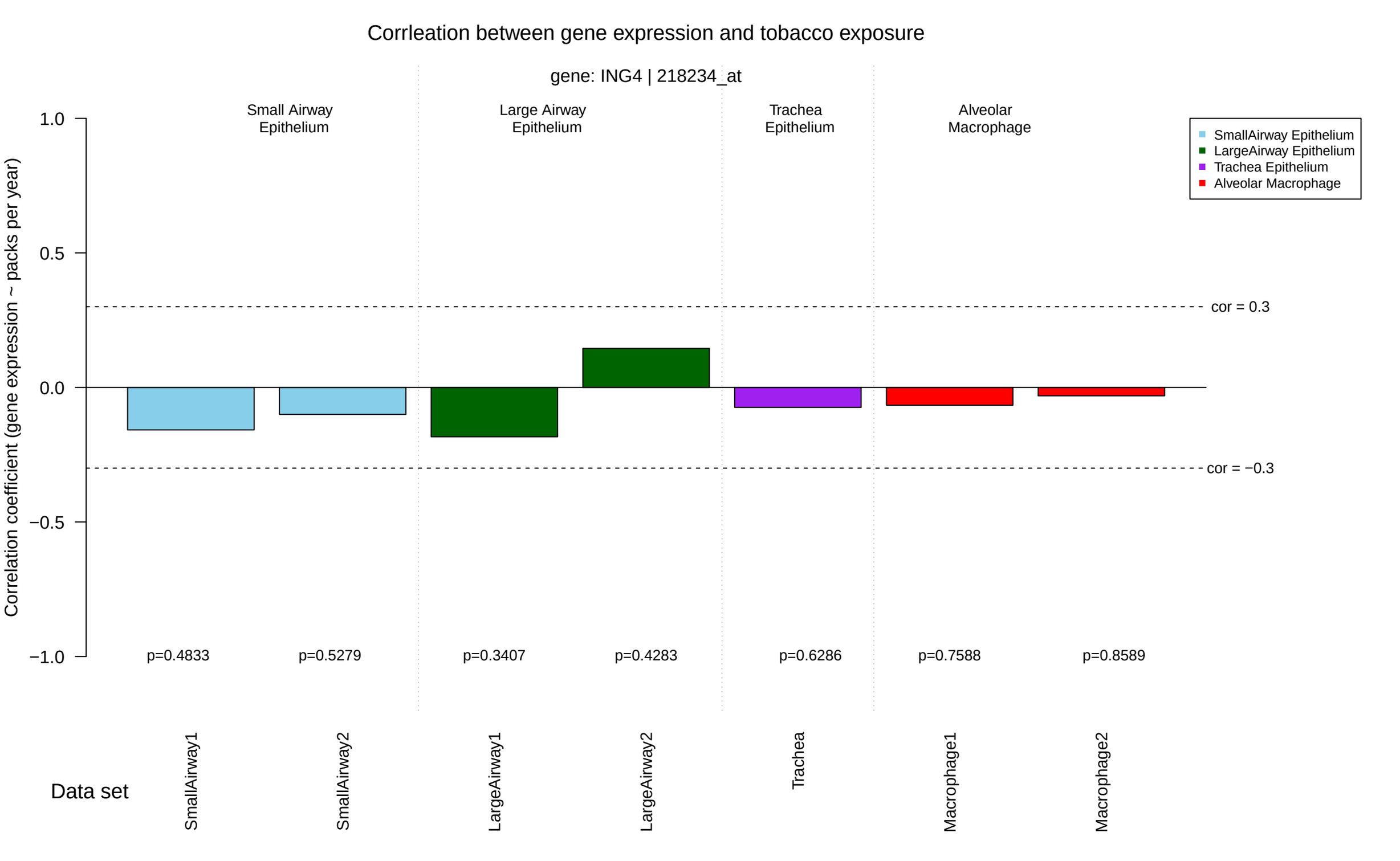

set: LargeAirway1 (p=0.2267)
ING4 | 218234_at
Expression level (log2)
6.0
7.0
8.0
9.0
20
30
40
50
60
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.5817)
ING4 | 218234_at
Expression level (log2)
6.0
7.0
8.0
9.0
20
30
40
50
60
Age
Smoker
non−smoker


set: Trachea (p=0.4669)
ING4 | 218234_at
Expression level (log2)
6.0
7.0
8.0
9.0
20
30
40
50
60
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1023)
ING4 | 218234_at
Expression level (log2)
6.0
7.0
8.0
9.0
20
30
40
50
60
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4836)
ING4 | 218234_at
Expression level (log2)
6.0
7.0
8.0
9.0
20
30
40
50
60
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ING4 | 218234_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4833
p=0.5279
p=0.3407
p=0.4283
p=0.6286
p=0.7588
p=0.8589
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2